

QITOUS VTOL Drone Project

AE 314: Systems & Control

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Cover photo: QITOUS VTOL prototype displayed during project presentation and demonstration activities.

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2025-12

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Abstract

This report documents the complete design, fabrication, integration, and validation process of the **QITOUS VTOL** aircraft developed for AE 314: Systems & Control. The platform was conceived as a hybrid VTOL system that combines quad-rotor hover capability with fixed-wing forward flight, targeting improved cruise efficiency while maintaining the operational flexibility of vertical takeoff and landing. The airframe was designed using a parametric CAD workflow in **Fusion 360** and manufactured primarily through **FDM 3D printing**, which enabled fast iteration, component replacement, and modular segmentation to satisfy printer build-volume constraints.

Material selection was treated as a system-level design variable. As mass growth emerged during prototyping, lightweight **PLA Aero** was adopted for large airframe shells and wing segments to reduce total weight while preserving adequate stiffness, whereas **PLA+** was reserved for higher-stress and handling-critical parts such as control surfaces and the nose section. During assembly, practical tolerance and handling issues were resolved through local sanding, adhesive repair, and sealing of small gaps, allowing the mechanical build to be completed without redesigning major structures late in the schedule.

From a controls and systems perspective, QITOUS implements an embedded stabilization loop based on a **Teensy 4.0** flight controller and an **MPU6050** inertial measurement unit. The software follows a classical feedback architecture: pilot commands are acquired and normalized, sensor signals are filtered to mitigate vibration-induced noise, attitude is estimated via a Madgwick-based 6-DOF algorithm, and control laws generate actuator commands through a mode-dependent mixer (hover versus cruise). Safety mechanisms (arming interlocks, throttle cut, and signal-range failsafe behavior) were incorporated to reduce risk during bench testing and progressive flight trials.

The report further summarizes observed performance trends, key implementation lessons (including vibration sensitivity and center-of-gravity coupling), and recommendations for future improvements such as transition validation, quantitative logging, and refined robustness against yaw drift in magnetometer-free operation.

Chapter 1

Introduction

1.1 Project Overview

The QITOUS VTOL is a 3D-printed VTOL aircraft developed in AE 314: Systems & Control. The platform integrates quad-rotor hover with fixed-wing forward flight to reduce power consumption during cruise compared to rotor-only flight. The project emphasizes end-to-end engineering integration, where mechanical design decisions directly impact sensing quality, actuator authority, and closed-loop stability.

1.2 Objectives

- Design and fabricate a VTOL-capable fixed-wing aircraft using 3D-printed components.
- Implement closed-loop attitude stabilization using an IMU and embedded control logic.
- Validate safe operation through bench tests and progressive flight tests.
- Document engineering decisions, challenges, iterations, and final outcomes.

1.3 Project Overview Poster

1.4 Key Specifications

Table 1.1: System specifications (reported/estimated)

Parameter	Value
Total mass	2.1 kg (reported)
Battery	14.8 V (4S) LiPo
Motors	2×920 KV and 2×935 KV brushless motors
Propellers	4× (9×4×3) glass-fiber propellers
Hover endurance	10.61 min (estimated)
Forward speed	8.6 m/s (estimated)
Estimated range	3.27 km (estimated)
Payload margin	1.1 kg (estimated)

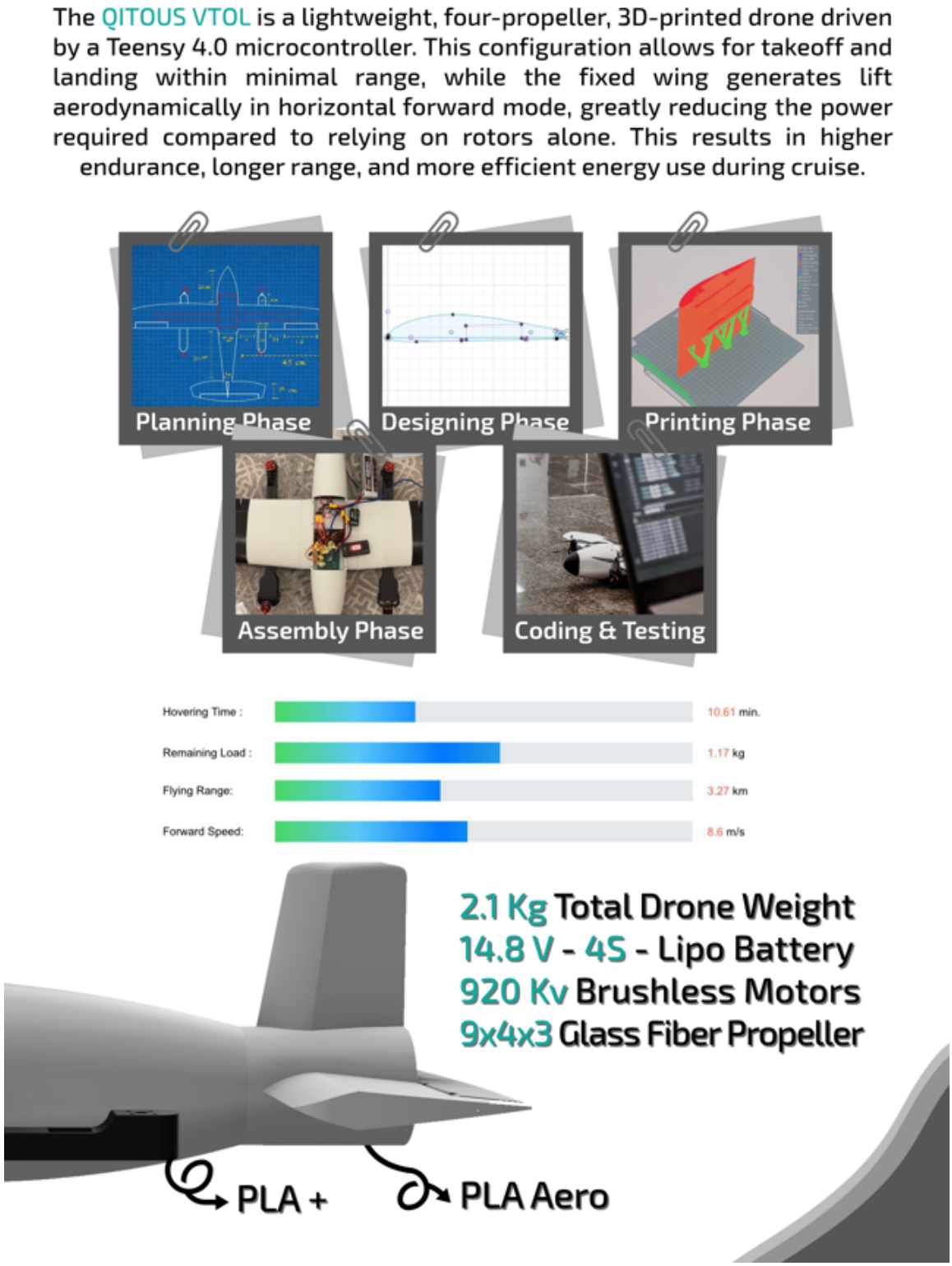


Figure 1.1: Project poster summary for the QITOUS VTOL, including configuration, key metrics, and development phases.

Chapter 2

Planning Phase

2.1 Requirements and Constraints

2.1.1 Functional Requirements

- Vertical takeoff/landing with stable hover.
- Forward flight supported by fixed-wing lift.
- RC command input with safety mechanisms (arming logic, throttle cut, failsafe).
- Modular design to enable iteration and maintenance access.

2.1.2 Design Constraints

- 3D-printer build volume (part segmentation and assembly features).
- Structural stiffness under thrust and landing loads.
- Electronics packaging constraints (battery, controller, receiver, ESCs, wiring).
- IMU sensitivity to vibration (mounting and balancing requirements).

2.2 How Did We Choose the Dimensions?

Dimensions were driven by performance targets, packaging, manufacturability, and aerodynamic considerations. In addition to hand calculations and packaging constraints, online sizing and stability checks were used to cross-validate wing planform proportions, arm spacing, and

component placement. The goal was to maintain a feasible center-of-gravity range, preserve adequate control authority in hover, and reduce avoidable redesign risk later in integration.

2.2.1 Early Concept and Dimensioning Sketch

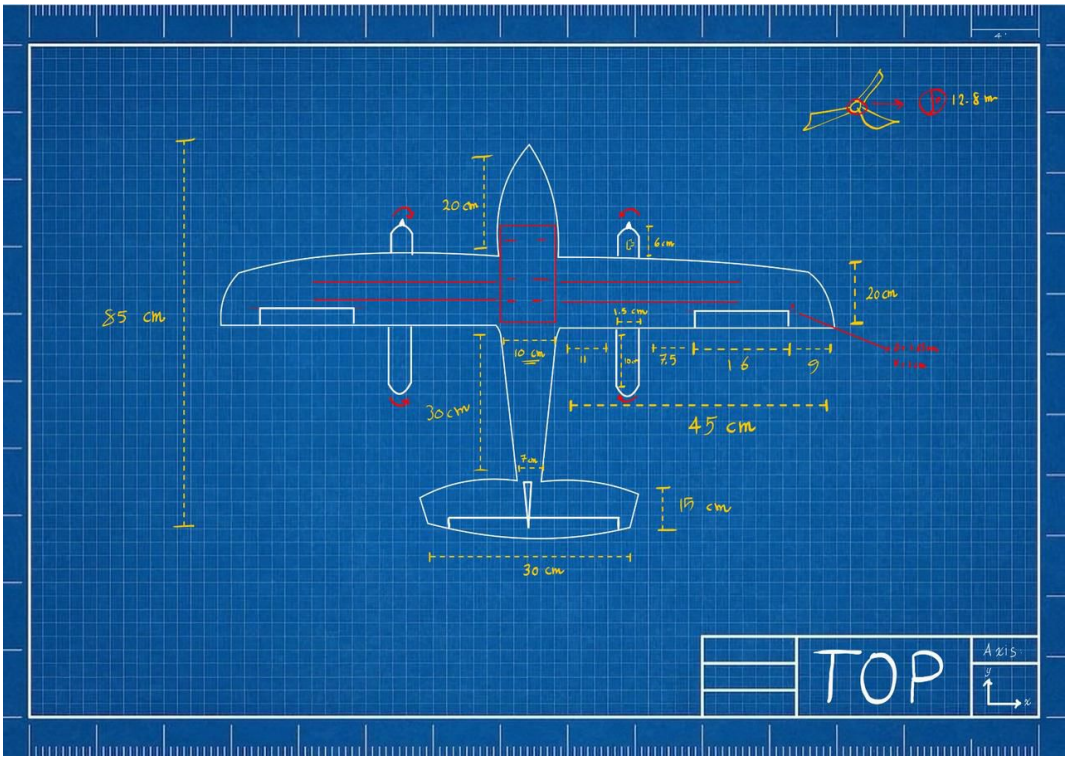


Figure 2.1: Early planning sketch showing the initial layout, principal dimensions, and component placement assumptions.

2.3 Why Four Propellers?

A four-propeller architecture was selected to maximize stability and controllability during hover. The quad configuration provides balanced thrust distribution and well-established control mixing for roll, pitch, and yaw. This reduces implementation uncertainty and supports a safer tuning workflow, especially during early flight trials where disturbances, CG offsets, and vibration effects are most pronounced.

2.4 Materials and Manufacturing

2.4.1 Why PLA+?

PLA+ was used primarily for the nose section and control surfaces. These parts experience repeated handling, potential impacts, and actuation loads. PLA+ provides improved stiffness and sharper dimensional detail than lightweight filaments, improving hinge fits, mounting tolerances, and robustness.

2.4.2 Why PLA Aero?

As the airframe mass increased during development, PLA Aero was introduced for major printed shells and wing segments to reduce weight while retaining sufficient stiffness. The material's foaming behavior reduces density while preserving a continuous outer skin, supporting lightweight prototypes without requiring excessively thin walls that can reduce durability.

2.4.3 Why 3D Printing?

- Rapid iteration and short design–test cycles.
- Complex geometries (integrated mounts, channels, alignment features).
- Low-cost customization compared to composite tooling or machining.

Chapter 3

Design Phase

3.1 Why Fusion 360?

Fusion 360 was selected due to its parametric workflow, assembly validation, and direct export to print-ready formats. Its collaborative features supported parallel work across team members and reduced iteration time. Parametric constraints also simplified late adjustments when packaging and CG targets required refinement.

3.2 CAD Workflow

3.2.1 Airframe Architecture

The airframe design was inspired by commercially available four-propeller VTOL drones. To meet printer build-volume constraints, the fuselage was split into multiple sections and the wing into segments. Motor holders were mirrored about the aircraft y-axis to preserve symmetry and thrust alignment. Ailerons and elevators were designed with integrated hinge mechanisms to ensure smooth rotation under servo loads and to reduce assembly complexity.

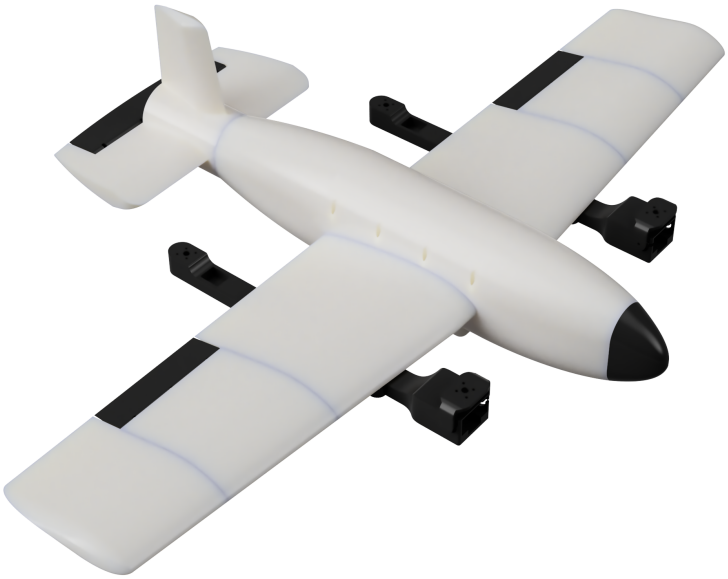


Figure 3.1: Fusion 360 airframe assembly view highlighting segmentation strategy, mounting interfaces, and integration features.

3.3 Wing Airfoil

The wing airfoil selected is **NACA 24012**, a NACA four-digit airfoil with 2% maximum camber located at 40% of chord and a 12% thickness-to-chord ratio. The selection supports manufacturability (printable thickness and internal support), low-speed lift behavior during transition and modest cruise, and practical handling qualities suitable for iterative flight testing.

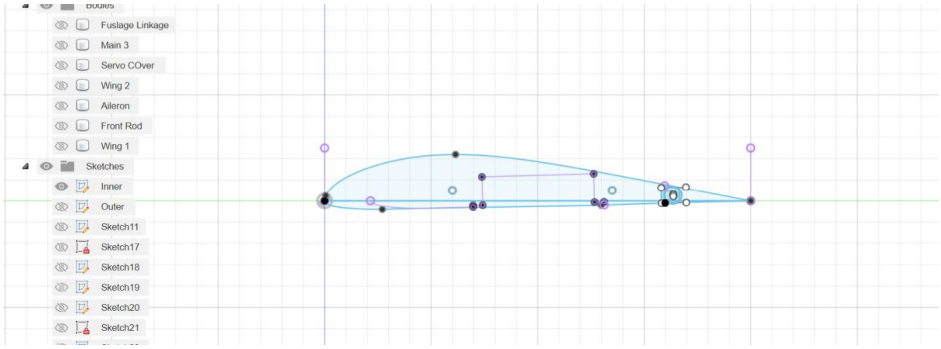


Figure 3.2: Airfoil profile used for the wing (NACA 24012), selected for low-speed lift performance and printable thickness.

Chapter 4

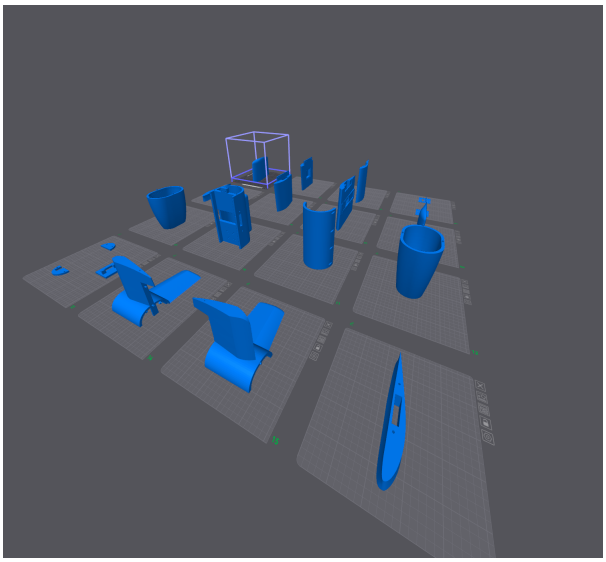
Printing Phase

4.1 Slicing Process (Orientation, Infill, Walls) and Rationale

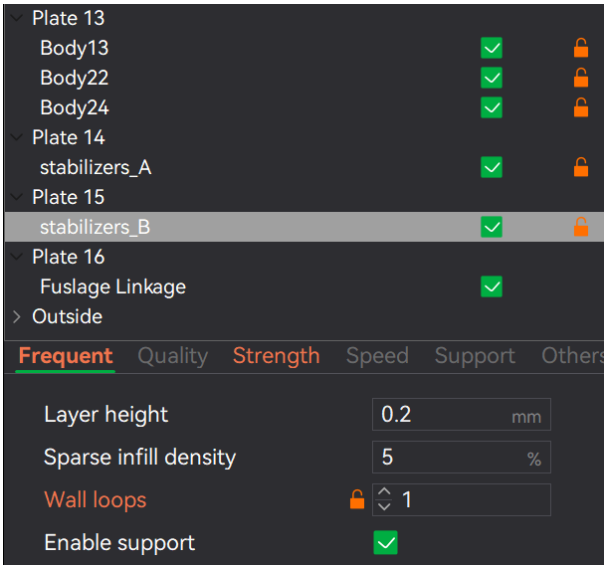
Slicing parameters were selected to balance stiffness, durability, and print time. The design intent was to ensure adequate bending and torsional stiffness for wing and fuselage shells while limiting mass and mitigating crack initiation at joints. Key considerations included:

- **Orientation:** reduce support contact on aerodynamic surfaces and align layer directions with primary load paths.
- **Infill:** provide internal shear support with minimal weight penalty and improved resistance to local buckling.
- **Walls/perimeters:** strengthen the outer skin and improve fatigue resistance under vibration.
- **Layer height:** improve dimensional accuracy at interfaces (hinges, mounts, joints) while keeping print time practical.

4.1.1 Slicing Screenshots



(a) Print orientation strategy used to reduce supports and align layers with expected load paths.



(b) Infill/perimeter configuration, including additional strength settings used in critical regions.

Figure 4.1: Slicing configuration examples used for QITOUS VTOL prints.

4.1.2 PLA Aero Printed Parts

4.1.3 Printers Used

The majority of structural components were produced using an FDM printer setup suitable for long-duration prints. Printer reliability and repeatability were essential because large wing and fuselage segments required consistent dimensional outcomes to avoid cumulative tolerance stack-up during assembly.

4.1.4 Problems Faced During Printing

Filament	Model	Support	Total
1	239.44 m 696.86 g	33.85 m 98.51 g	273.28 m 795.37 g

Total cost: 35.78

Time Estimation	
Plate 1	2h33m
Plate 2	3h38m
Plate 3	6h57m
Plate 4	43m46s
Plate 5	1h29m
Plate 6	3h40m
Plate 7	6h56m
Plate 8	2h30m
Plate 9	3h43m
Plate 10	9h13m
Plate 11	3h17m
Plate 12	6h3m
Plate 13	47m15s
Plate 14	5h54m
Plate 15	5h57m
Plate 16	1h29m
Total	2d16h49m

Figure 4.2: PLA Aero printed components used to reduce airframe mass while maintaining sufficient stiffness for assembly and flight testing.

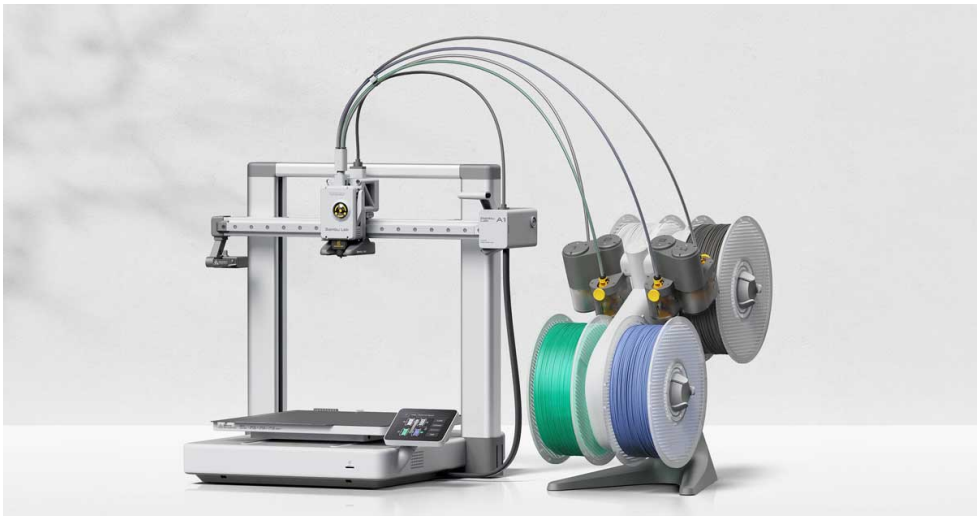


Figure 4.3: Representative FDM 3D printer configuration used for manufacturing large airframe segments and structural components.

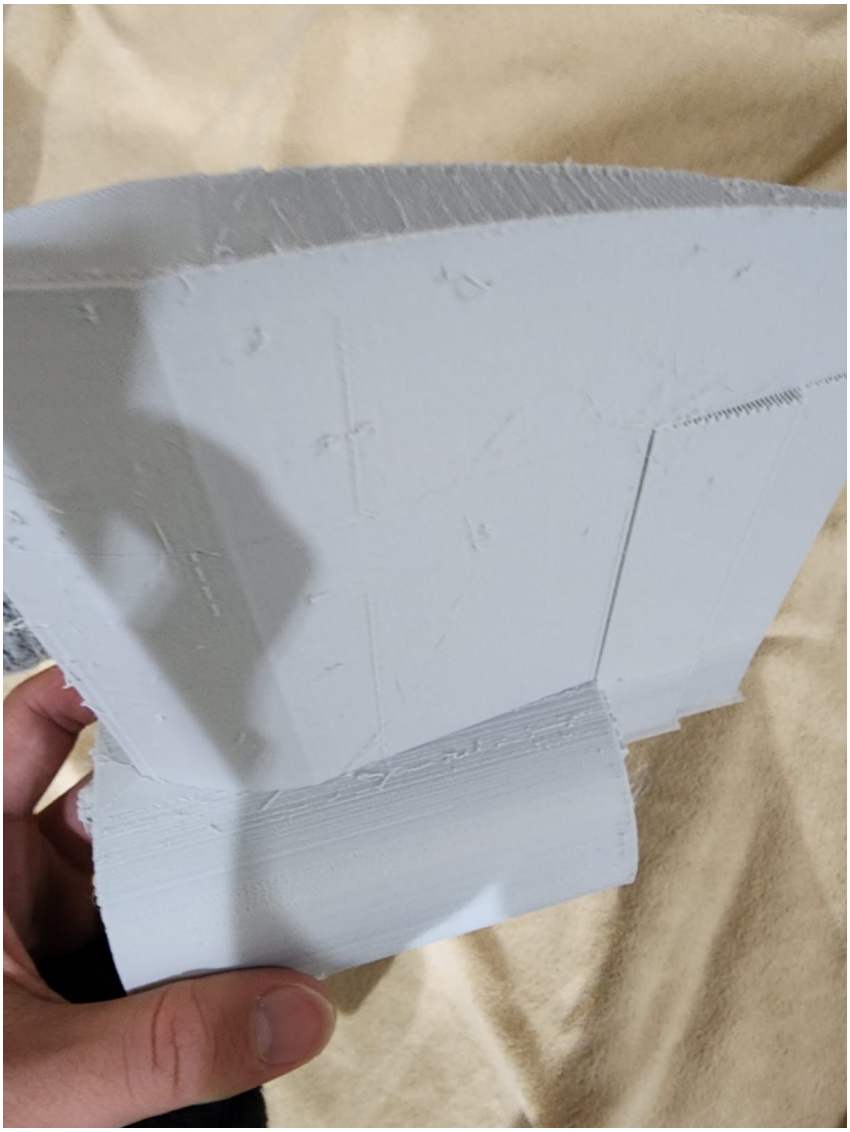


Figure 4.4: Representative print issues observed during manufacturing (e.g., warping, support defects, and tolerance mismatch) and resulting corrective actions.

Chapter 5

Assembly Phase

5.1 Mechanical Assembly



Figure 5.1: Mechanical assembly stage showing airframe integration, joint alignment, and internal packaging progression.

5.1.1 Mechanical Assembly Challenges and Solutions

During the mechanical assembly phase, several practical challenges were encountered that required on-site modifications and corrective actions. Minor dimensional mismatches between printed components led to difficulties in fitting parts together accurately; this issue was resolved by carefully sanding the affected surfaces until proper alignment and fit were

achieved. Additionally, a few structural components were damaged during handling and assembly. These broken parts were successfully repaired using industrial super glue, restoring sufficient mechanical strength for continued use. Furthermore, unintended holes and gaps appeared in some support structures due to manufacturing tolerances. To address this, excess material from the same support parts was locally melted and used to fill and seal the holes, ensuring improved structural integrity and rigidity. Overall, these corrective measures allowed the mechanical assembly to be completed successfully despite initial fabrication and assembly limitations.

5.2 Electrical Integration (Wiring and Mounting)

To ensure stable control and safe operation, wiring was routed to minimize strain, prevent connector loosening under vibration, and reduce the likelihood of intermittent signal loss. Power distribution and ESC signal routing were organized to simplify troubleshooting during bench testing and tuning.

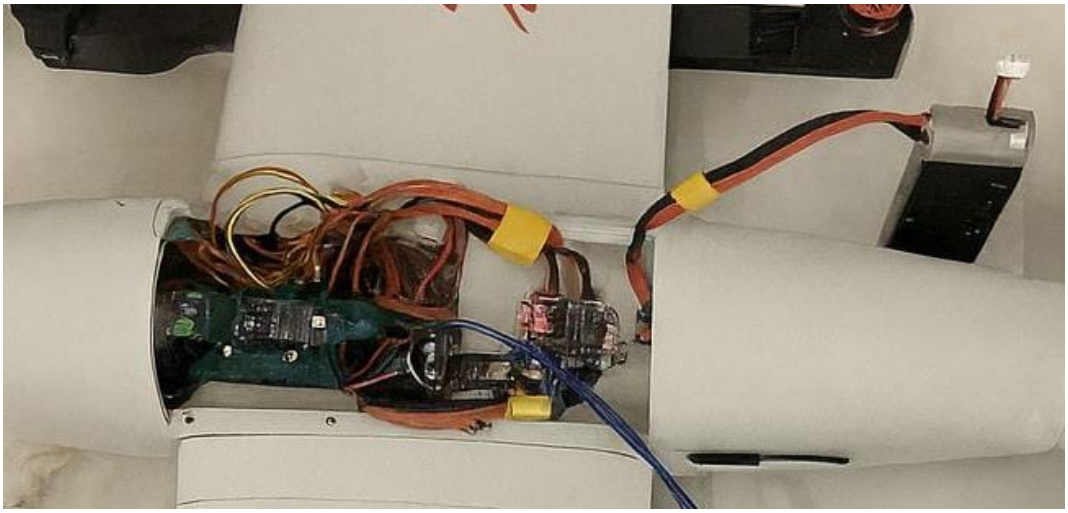


Figure 5.2: Final wiring layout and routing, including controller interconnects, ESC signal wiring, and power distribution integration.

5.3 Motor Vibration Issues and Solutions

Motor-induced vibrations can corrupt IMU measurements, inject noise into the feedback loop, and reduce stability margins. Mitigation steps included:

- Propeller balancing and inspection for damage.

- Reinforced motor mounts and correct fastener tightening.
- IMU isolation (foam damping), placement near the CG, and cable strain relief.

Chapter 6

Coding & Testing Phase

6.1 Important Note on Source Code Attachment

The full firmware source code for this project will be attached to the submitted report as a separate file (e.g., `QITOUS_Teensy_7Dec.zip`). For readability and report scope, code snippets are intentionally omitted.

6.2 Software Architecture (High-Level)

- RC input acquisition and normalization
- IMU sampling and filtering
- Attitude estimation (Euler angles)
- PID control loops (angle mode / yaw rate)
- VTOL mixing logic (hover vs. cruise)
- Safety: arming, throttle cut, failsafe handling

6.2.1 Coding & Testing

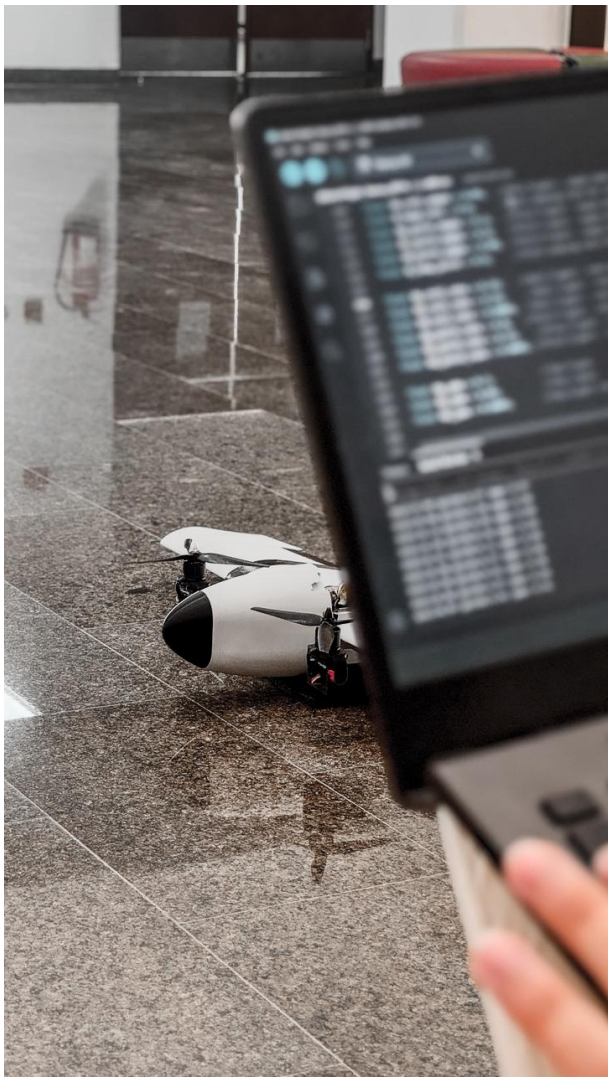


Figure 6.1: Firmware development and bench-testing workflow used during sensor verification, actuator checks, and control tuning.

6.3 Control and Systems Theory Background (Applied)

The QITOUS VTOL is a multi-input multi-output (MIMO) system whose effective dynamics vary between hover and forward-flight modes. In hover, the controlled outputs are roll, pitch, and yaw, while thrust primarily determines vertical acceleration. Around a trim condition and for small angles, the rotational dynamics can be approximated by linearized models, enabling classical reasoning about stability margins, disturbance rejection, and sensitivity to noise and delays.

6.3.1 Closed-Loop Feedback Structure

The implemented autopilot follows a negative-feedback structure: filtered sensor measurements are compared against commanded references, and the error is mapped to actuator commands through the control law and mixer. A common representation of the local dynamics is:

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t),$$

where x denotes states, u actuator commands, and y measured outputs. In practice, performance is limited by sensor noise, unmodeled dynamics, time delay, and actuator saturation, which must be considered during tuning and validation.

6.3.2 PID Control and Practical Stabilization

Roll and pitch are stabilized using PID-type control in angle-mode, while yaw is stabilized using rate feedback. The control objectives include: (i) fast response without overshoot, (ii) adequate damping against disturbances, and (iii) robustness against vibration-induced measurement noise. Anti-windup measures were required; therefore, integrators were gated at low throttle and output commands were saturated to safe PWM limits.

6.3.3 Filtering and Sensor Fusion

Vibration introduces high-frequency components in accelerometer and gyroscope signals. Low-pass filtering reduces noise and improves robustness, at the cost of phase lag. Attitude estimation is computed using a Madgwick-based 6-DOF filter combining gyroscope integration with accelerometer gravity referencing. Without a magnetometer, yaw drift is expected; therefore, smoothing was introduced to reduce short-term jitter, while acknowledging that long-term heading drift cannot be fully eliminated without absolute heading correction.

6.3.4 Mode Switching and Mixer Allocation

Hover and cruise modes use different actuator allocations; consequently, mode switching changes the effective mapping from controller outputs to actuator commands. For safety, testing was conducted progressively to ensure stable hover operation before evaluating forward-flight and any transition behavior.

6.4 Implemented Code Modifications (Latest Version)

6.4.1 Radio & Flight-Mode Logic

- Receiver mode set to **PPM** (`#define USE_PPM_RX`).
- Channel mapping and normalization clarified for FS-i6X (AETR): CH3 throttle, CH1 roll, CH2 pitch, CH4 yaw.
- Flight-mode switch: CH6 low = Hover, CH6 high = Cruise.

6.4.2 IMU, Frames & Filtering

- MPU6050 over I^2C with 250 dps / 2 g scales.
- Low-pass filtering with stronger yaw-rate filtering.
- Wrap-safe yaw smoothing applied after Euler extraction.
- IMU axis alignment fix implemented in calibration.

6.4.3 Mixer, Servos & Surfaces

- Dual-mode mixer for hover versus cruise allocation.
- Servo direction correction and tightened PWM constraints.
- KV compensation applied after motor assignment and before saturation.

6.4.4 Safety, Arming & Throttle Cut

- Arming requires CH5 low and throttle low.
- Throttle cut forces motors to 1000 μs when CH5 high or not armed.
- Failsafe defaults when RC readings exceed valid bounds.

6.5 Testing Plan and Validation Evidence

6.5.1 Bench Tests (No Propellers)

- Verify IMU axes and sign conventions.
- Verify RC channel mapping and flight-mode switch.
- Verify arming/throttle cut/failsafe behavior.
- Verify motor order and direction (without props).

6.5.2 Flight Tests (Progressive)

- Short hover tests at low altitude.
- Gain tuning to reduce oscillation and improve response.
- Cruise-mode checks and transition experiments (if performed).



Figure 6.2: Flight testing evidence and gain-tuning iterations (replace with selected video frames if improved clarity is desired).

Chapter 7

Results and Discussion

7.1 Performance Summary

7.2 Key Observations

- Small CG offsets significantly affected stability and increased tuning effort, confirming strong mechanical–control coupling.
- Achieving a robust balance between K_p, K_d, and K_i required iterative tuning to avoid oscillations while maintaining responsiveness.
- Control surfaces and the tilt mechanism operated reliably during testing, supporting repeatable mechanical response.
- Vibration mitigation improved attitude estimation consistency and reduced noise in the closed-loop response.

7.3 Main Challenges

- CG sensitivity highlighted the need for careful packaging design and mass distribution control.
- Persistent vibration sources required combined mechanical mitigation and software filtering.
- Yaw drift was observed due to the absence of magnetometer-based absolute heading correction.

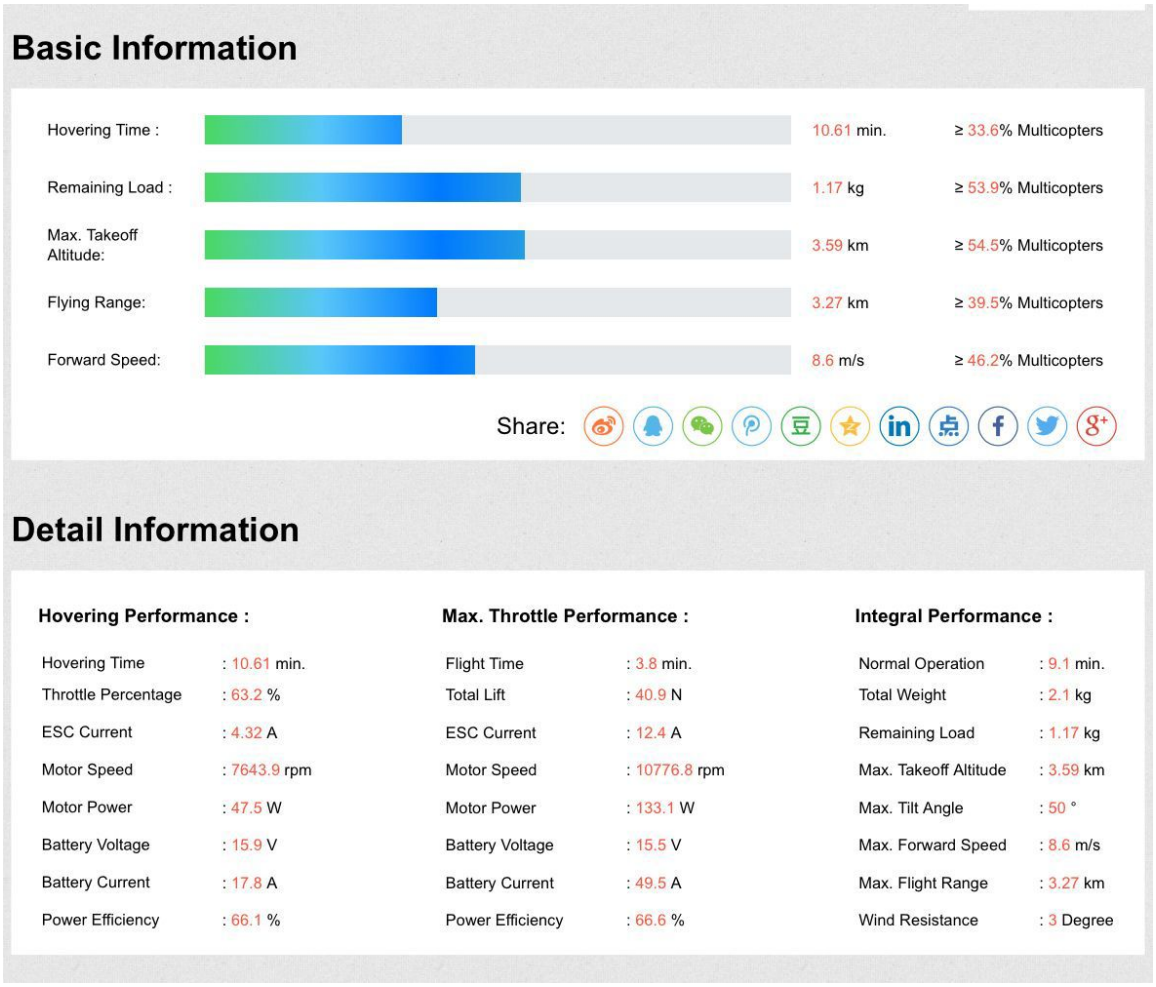


Figure 7.1: Estimated performance summary including hover endurance, payload margin, range, and forward speed.

7.4 Equipment Losses and Lessons Learned

7.4.1 Laptop Battery Damage During Bench Debugging

A laptop used during bench work suffered battery damage consistent with an electrical fault during testing activities. This motivates stronger isolation practices: correct power-domain separation, protective USB isolation/hubs, controlled power-up sequences, and verified grounding strategies.

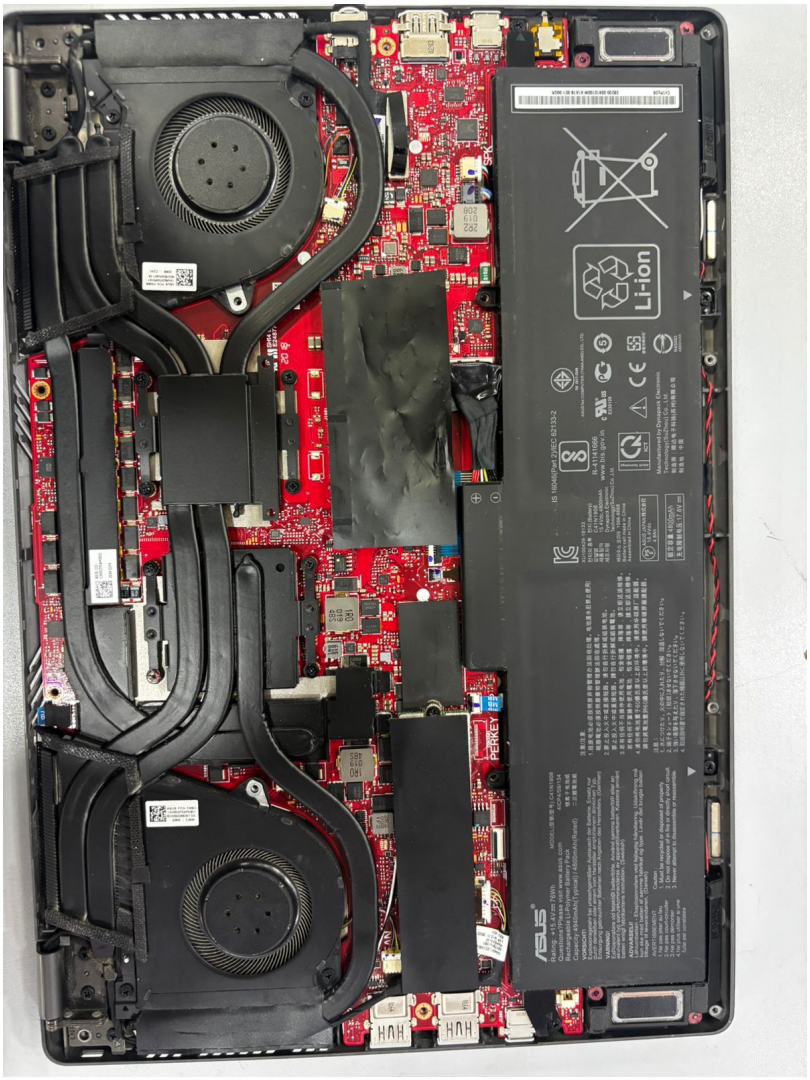


Figure 7.2: Laptop internal damage incident recorded during development; motivates stricter electrical isolation and protection practices during bench testing.

7.4.2 Motor Winding/Wire Damage

One motor experienced winding/wire damage (stripped coil wires), preventing reliable operation. Potential contributors include handling damage, vibration, thermal stress, and repeated high-load testing. This supports the need for conservative current limits, routine inspection, and improved strain relief and mount stiffness.



Figure 7.3: Motor damage observed during development (winding/wire degradation), highlighting the importance of inspection routines and vibration mitigation.

Chapter 8

Materials and Hardware Components

8.1 3D-Printing Materials

Table 8.1: Printing materials used

Material	Primary Use	Quantity
PLA+	Nose section and control surfaces	240 g
PLA Aero	Weight-reduced airframe sections	810 g
Foam / damping	IMU isolation and vibration reduction	as required
Steel push rod (1.2 mm)	Links control surfaces	5 rods

8.2 Hardware & Electronics

Table 8.2: Hardware and electronics list

Component	Count	Notes
Radio transmitter	1	Ground control unit
Radio receiver	1	PPM receiver input
Microcontroller (Teensy 4.0)	1	Main controller
IMU (MPU6050)	1	6-DOF attitude estimation
Brushless motors	4	Four-propeller configuration
ESCs	4	One per motor
Propellers	4	9×4×3 glass-fiber
Battery (4S LiPo)	1	14.8 V nominal
BEC	1	Power regulation for electronics/servos
Servos	5	Tilt + control-surface actuation
Wiring/connectors	as required	Integration and assembly

Chapter 9

Conclusion

9.1 Conclusion

The QITOUS VTOL project successfully delivered a complete engineering cycle from conceptual planning and sizing through CAD design, additive manufacturing, mechanical/electrical integration, embedded software development, and progressive validation. The modular 3D-printed architecture enabled rapid iteration under build-volume constraints, and the final structure demonstrated that FDM manufacturing can support a functional VTOL prototype when segmentation, joints, and mounting interfaces are engineered carefully. Material selection was a major contributor to feasibility: PLA Aero enabled meaningful mass reduction for large airframe elements, while PLA+ preserved strength and dimensional quality for parts exposed to handling and actuation loads.

From a systems and control standpoint, the project validated practical closed-loop stabilization on a real aerial platform under non-ideal conditions. Sensor noise, vibration coupling, actuator saturation, and mode-dependent mixing significantly influenced achievable performance and the effort required to tune gains. The final firmware improvements—especially the clarified RC mapping, dual-mode mixer, stronger yaw-rate filtering, yaw smoothing, and conservative safety logic—improved robustness during bench testing and reduced risk during flight trials. The experimental results reinforced that stable VTOL operation is not determined by control design alone; it depends on coordinated mechanical design (CG placement, stiffness, vibration), sensing quality, actuator health, and safe integration practices.

In summary, QITOUS VTOL provided a functional platform for applying classical control concepts (feedback stabilization, filtering, and robustness considerations) in a real embedded aircraft environment. The project outcomes and documented lessons establish a strong foundation for further improvements, particularly for transition validation, heading stability, and quantitative performance characterization.

9.2 Future Work

- Implement structured logging (IMU, setpoints, PID terms, motor outputs) to support quantitative tuning and system identification.
- Develop and validate a safer transition strategy (blending, gain scheduling, and bounded rate limits) before attempting extended forward flight.
- Improve yaw observability by adding magnetometer/heading correction (or alternative reference), reducing long-term drift.
- Reduce vibration at the source through improved prop balancing, stiffer mounts, and refined isolation placement near the CG.
- Optimize mass distribution and packaging to reduce CG sensitivity and simplify repeatable trimming and tuning.

Appendix A

Attached Materials

The following files are submitted as attachments with this report:

- Source Code (ZIP): `QITOUS_Teensy_7Dec.zip`
- Documentation PDFs: parts list and VTOL documentation files